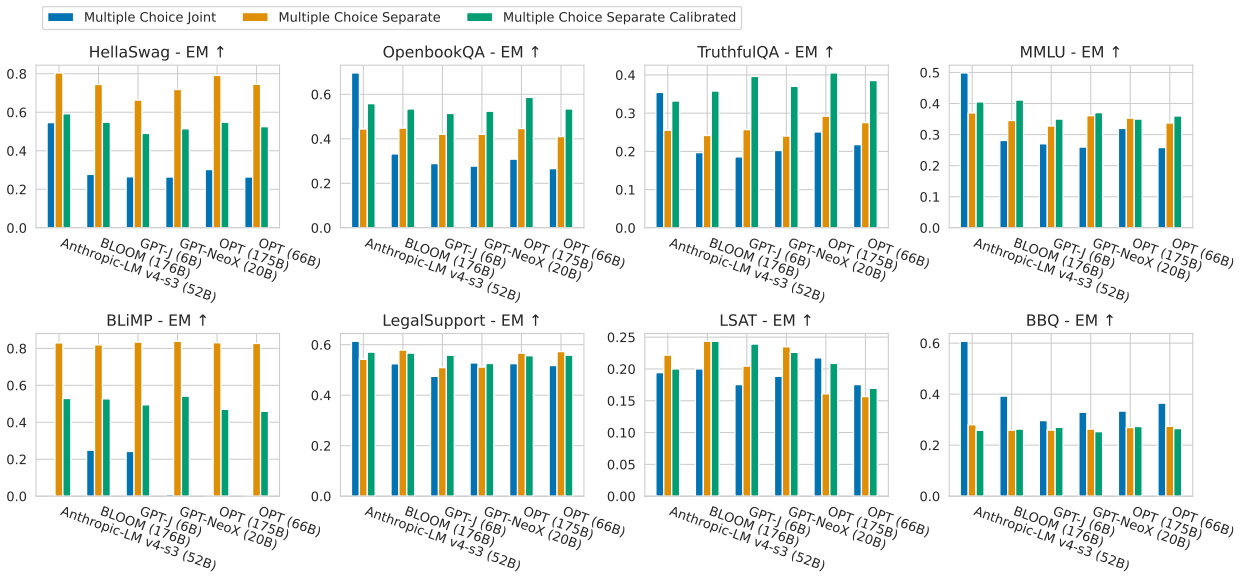

Figure 33: **Multiple-choice adaptation.** For each adaptation method (joint, separate, and separate calibrated), we compare models across scenarios.

Formulation of multiple choice scenarios. Beyond the details of the prompt, we can conceptually imagine different ways to make use of the language interface to perform the same underlying scenario. As we discuss in Appendix J, in the case of multiple choice scenarios, we could provide the model with all of the answer choices and task it with selecting the correct choice (*joint*). Or we could provide the model with each answer choice as a separate query of the model, and see which answer choice is assigned higher probability (*separate*). And we could further calibrate the model-assigned probabilities by the probability the model assigns to the corresponding answer choice when presented standalone (*separate-calibrated*).

The results for these multiple choice scenarios as a function of this choice can be found at `https://crfm.stanford.edu/helm/v0.1.0/?group=ablation_multiple_choice`, and they are visualized in Figure 33. We observe that the method that maximizes accuracy is largely determined by the scenario, whereas it is generally consistent across models for a given scenario. Taking **HellaSwag** as a case study, for all six models, the methods follow *separate* > *separate-calibrated* > *joint*. This choice can be very consequential for the model accuracy: for OPT (175B), the accuracies are 79.1%, 54.8%, and 30.2% respectively. Further, the trends are entirely inverted for calibration error, as may be expected. So, overall for **HellaSwag**, across both desiderata and for all six models, there is clear and strict Pareto dominance in adaptation methods of *separate* > *separate-calibrated* > *joint*. In the case of **HellaSwag**, given the dataset is designed as completions of an incomplete textual sequence, the preference for the *separate* adaption method over the *joint* adaptation method is fairly intuitive and precisely aligns with the empirical findings. However, we do note the methods pattern differently for specific models in terms of efficiency. In particular, the *separate* methods require a query for each answer choice, whereas the *joint* method requires a single, much longer (especially due to the inclusion of in-context examples) query. This manifests in lower denoised inference times for *separate* over *joint* for BLOOM (176B) (0.075s vs. 0.271s), but the opposite trend for OPT (175B) (0.71s vs. 0.187s), despite the fact the models are evaluated on the same infrastructure and are near-identical in overall parameter count.

For the other scenarios, we do not observe as model-agnostic preferences over the adaptation methods. In general, for accuracy, we find *separate-calibrated* > *separate* > *joint* for **OpenBookQA**, **TruthfulQA**, and **MMLU** for five of the six models. The consistent exception is Anthropic-LM v4-s3 (52B) where the preference is consistently *joint* > *separate-calibrated* > *separate*. We find this to be quite striking: the adaptation method that elicits the most accurate behavior from Anthropic-LM v4-s3 (52B) elicits the least accurate behavior from the other five models. These gaps are significant as well: on **OpenBookQA**,

presenting accuracies in the order (*separate-calibrated*, *separate*, *joint*), we see (58.6%, 44.6%, 30.9%) for OPT (175B), whereas for Anthropic-LM v4-s3 (52B) we see (55.8%, 44.4%, 69.6%). That is, if OPT (175B) and Anthropic-LM v4-s3 (52B) were compared using the *separate-calibrated* adaptation method (or the *separate* method), they would achieve near-equal accuracies (within 3%), but they are almost 40% apart when using the *joint* method, and 12% apart when comparing the per-model accuracy-maximizing method. This raises fundamental questions about the appropriate way to compare across models and the sense in which uniform standardization is fair; we encourage future work to more extensively look into this and develop clearer best practices.

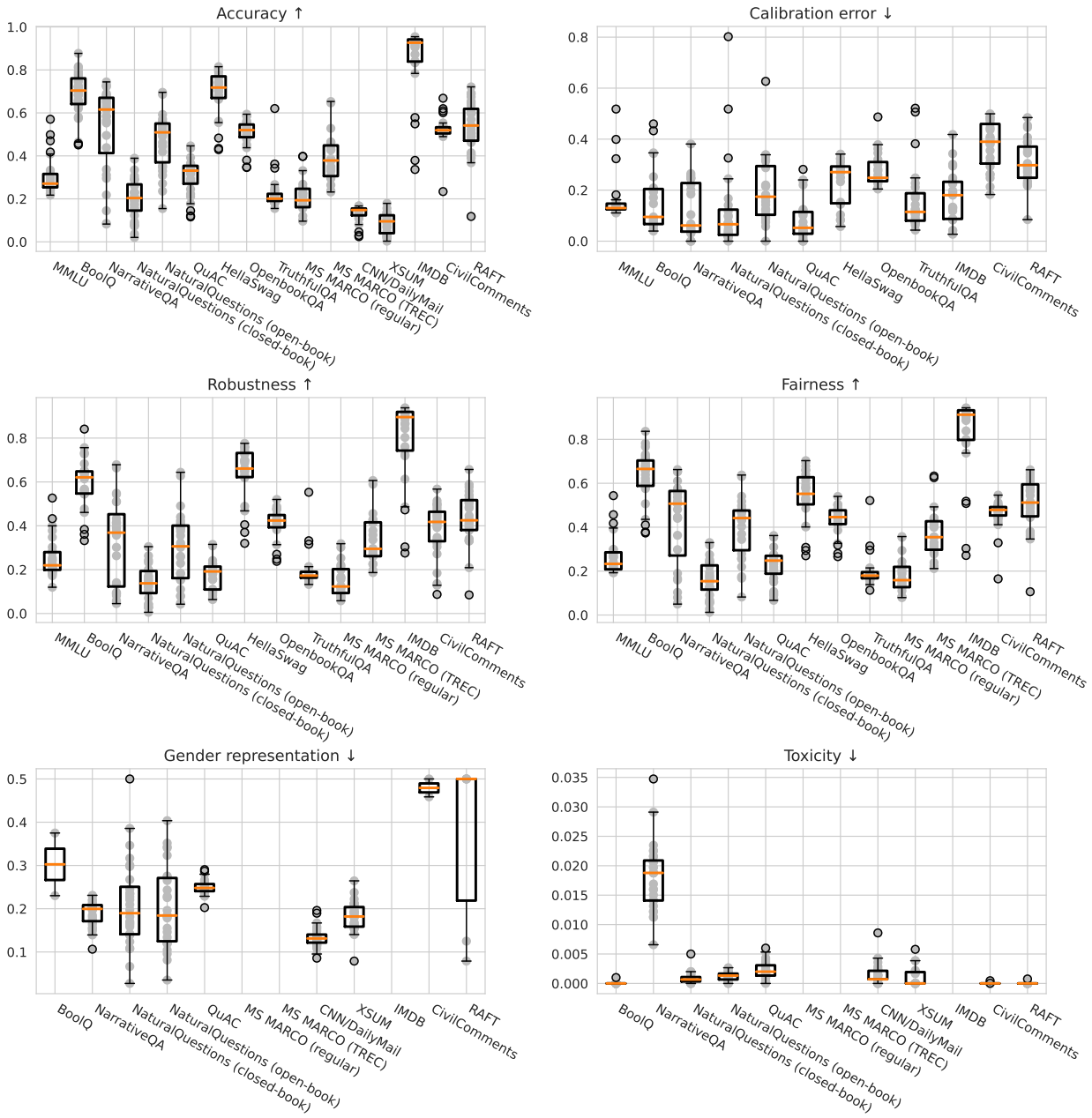

Figure 34: **Metric spread for core scenarios.** Metrics for every model on every core scenario as a means for indicating the spread on a per-metric basis.

8.3 Task-specific results for core scenarios

Since we organize the 16 core scenarios by the broader task, we highlight findings at the task level. To provide a sense of the spread in accuracies for each of these scenarios, we provide Figure 34. The results we provide here are highlighting specific trends/phenomena, though we encourage interactively looking at the quantitative results and underlying model behaviors.

Question answering. To further explore the results for this task, see `https://crfm.stanford.edu/helm/v0.1.0/?group=question_answering`. For the 9 question answering scenarios, we see significant heterogeneity across the results. However, in terms of accuracy, what is very consistent is that text-davinci-002